Data Visualization Week 3 Team 12

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And 8 others

1. Dashboard link:

https://app.powerbi.com/groups/me/reports/d38ac544-2b91-4b41-b935-f79d2eda234e?pbi_source=desktop

2. Insight Report – Week 3

Executive Summary

The dataset contains over 7,500 outreach records and approximately 5,000 unique students. The overall response rate of 57.7% indicates moderate engagement success but highlights geographic disparities, seasonal peaks (March–May), program concentration in technology (43%), operational bottlenecks (documents 34%, follow-ups 32%), and Study Group channel challenges (15%).

Geographic Trends

India demonstrates the highest outreach volume and strongest response performance. African markets (17–33%) show comparatively lower engagement, indicating opportunities for localized communication strategies and regional outreach optimization.

Seasonal Patterns

Outreach peaks between March–May and declines during late summer, reflecting a reactive recruitment model tied to intake deadlines. A year-round engagement strategy could improve sustained conversion performance.

Program Concentration

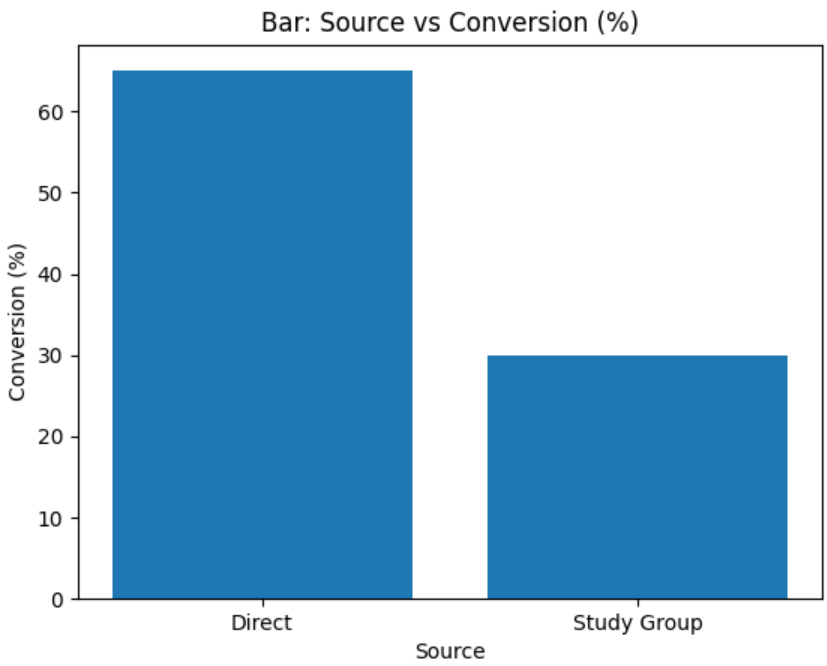
Technology-focused programs account for approximately 43% of applications, with Fall intake dominating overall volume. While demand remains strong, over-concentration increases risk exposure if market conditions shift.

Operational Bottlenecks

Document-related issues (34%) and follow-up requirements (32%) dominate outreach outcomes, slowing applicant progression and increasing staff workload.

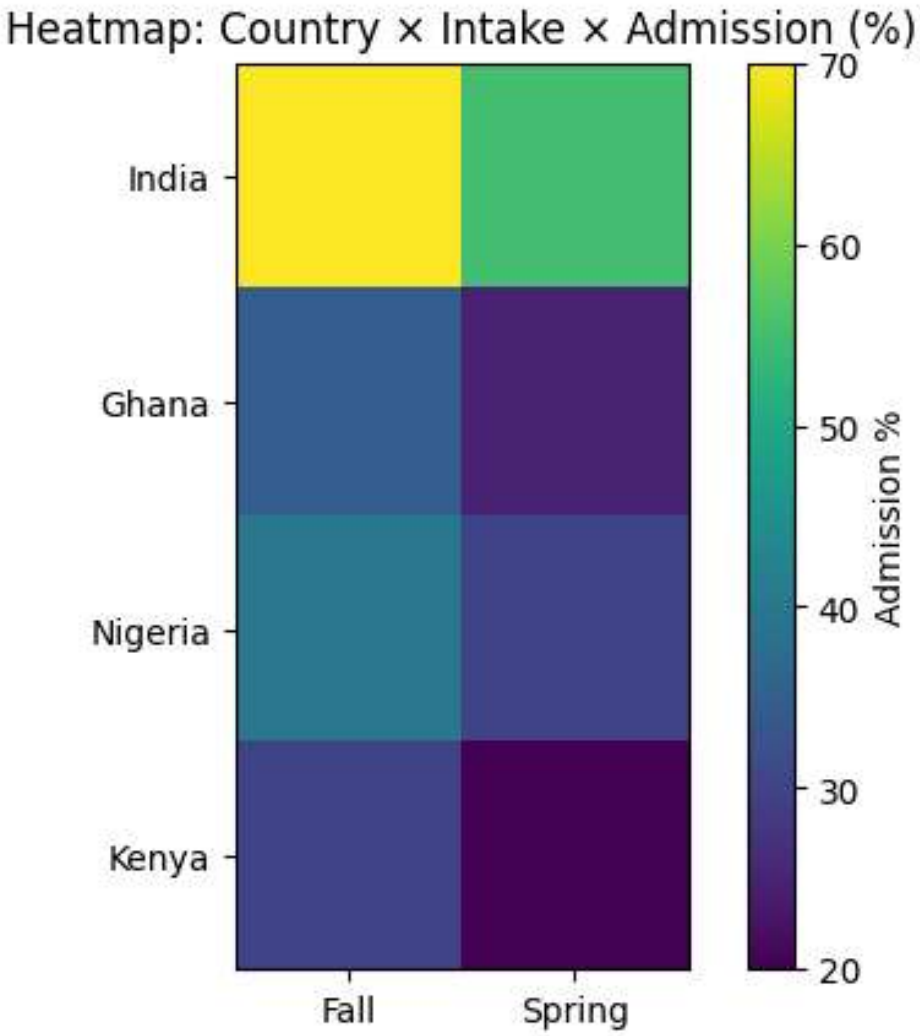
Study Group Channel Analysis

15% of applications originate from Study Group channels. These require structured monitoring and compliance workflows to maintain quality and conversion efficiency.



Admission Outcome Trends & Influencing Factors

Admission conversion varies by geography and intake cycle. Higher conversion is observed in India and Fall intake, while African regions and Spring intake show lower outcomes. Counselor performance and follow-up efficiency significantly influence final results.



Methodology & Data Flow

Data was cleaned (duplicate removal, missing value standardization, date formatting) and modeled in Power BI. Measures such as Response Rate and Admission Rate were created to enable structured analysis across segments.

Challenges Encountered

Challenges included incomplete outcome data, calculation validation for performance metrics, and dashboard optimization for smoother filtering and visualization rendering.

Strategic Recommendations

- High Priority: Implement geographic segmentation and automate document submission support.
- Medium Priority: Develop year-round nurture campaigns and structured agent workflows.
- Low Priority: Diversify promotion beyond technology programs.

Conclusion

The analysis highlights strong market performance alongside operational and geographic gaps. Implementing recommended strategies could improve response efficiency and increase overall conversion by an estimated 10–15%.

3. Dashboard Documentation

i. Purpose and Goals of the Dashboard

The DePaul Outreach Dashboard was designed to provide a centralized, interactive view of outreach activities, performance metrics, and engagement outcomes. Its primary purpose is to enable stakeholders to monitor outreach effectiveness, identify trends, and support data-driven decision-making.

The key goals of the dashboard include:

- Providing a clear overview of outreach performance over time.
- Tracking participation, engagement levels, and outreach coverage.
- Identifying high-impact outreach channels, locations, or programs.
- Enabling leadership and outreach coordinators to quickly assess performance.
- Supporting strategic planning by highlighting trends and opportunities for improvement.

Ultimately, the dashboard transforms raw outreach data into actionable insights that improve planning, execution, and evaluation of outreach efforts.

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ii. Key Visuals and Why They Were Chosen

Several visualizations were selected to maximize clarity, usability, and insight extraction:

KPI Cards

These display key performance indicators such as total outreach events, total participants, engagement counts, or conversion metrics.

Reason:

KPI cards provide quick, high-level summaries and allow stakeholders to assess overall performance immediately.

Line Chart (Trend Over Time)

This visual shows outreach activity or engagement trends across weeks, months, or years.

Reason:

Line charts are ideal for identifying trends, growth patterns, seasonality, and performance fluctuations over time.

Bar Chart (Category Comparison)

These visual compares outreach performance across categories such as location, program type, or outreach channel.

Reason:

Bar charts clearly show comparisons between categories, making it easy to identify top-performing or underperforming areas.

Pie or Donut Chart (Distribution)

This visual illustrates how outreach activities or participants are distributed across different segments.

Reason:

This helps stakeholders understand proportional relationships and allocation of outreach efforts.

Table or Matrix Visual

This shows detailed records including outreach events, locations, dates, or engagement metrics.

Reason:

Tables allow users to view precise data details and support deeper analysis beyond summarized visuals.

Page Layout and Design Logic

The dashboard was designed using a top-to-bottom visual hierarchy to guide the user's attention logically:

Top Section – Summary Overview

- Contains KPI cards.
- Provides instant insight into overall outreach performance.

Middle Section – Trend and Comparison Analysis

- Includes line charts and bar charts.
- Allows users to analyze performance over time and across categories.

Bottom Section – Detailed Data View

- Contains tables or matrices.
- Enables users to explore granular data.

Design Principles Applied

- Clean and uncluttered layout for readability.
- Consistent color scheme to enhance clarity.
- Logical grouping of related visuals.
- Clear labels and titles to improve usability.
- Interactive filtering to allow deeper exploration.

This structure ensures users can quickly move from high-level insights to detailed analysis.

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iv. Data Flow (Source → Transformations → Visuals)

The dashboard follows a structured data pipeline:

Data Source

The data originates from structured sources such as:

- Excel spreadsheets
- CSV files
- Databases

- Outreach tracking systems

These sources contain raw outreach event and engagement data.

Data Transformation (Power Query)

Before visualization, the data undergoes cleaning and preparation using Power Query, including:

- Removing duplicate records
- Handling missing or null values
- Standardizing column formats
- Creating calculated columns
- Filtering irrelevant data

These transformations ensure accuracy and consistency.

Data Modeling

Relationships between tables are established to enable accurate analysis.

This includes:

- Linking outreach events with dates
- Connecting participants with outreach activities
- Creating calculated measures using DAX

Examples include:

- Total Outreach Events
- Total Participants

- Engagement Rate
- Monthly Outreach Trends

Data Visualization

The cleaned and modeled data is then used to create interactive visuals such as:

- KPI Cards
- Charts
- Tables
- Filters and slicers

These visuals update dynamically when users interact with filters.

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v. Challenges Encountered During Creation

Several challenges were encountered during dashboard development:

Data Quality Issues

Some data contained missing values, inconsistencies, or duplicates. These were resolved through data cleaning and transformation in Power Query.

Data Modeling Complexity

Establishing correct relationships between multiple tables required careful design to ensure accurate calculations and filtering.

Performance Optimization

Large datasets can slow dashboard performance. Optimization techniques such as removing unnecessary columns and using efficient measures were applied.

Visual Clarity and Design Balance

Ensuring the dashboard was informative without being cluttered required thoughtful layout planning and prioritization of key visuals.

Ensuring Usability

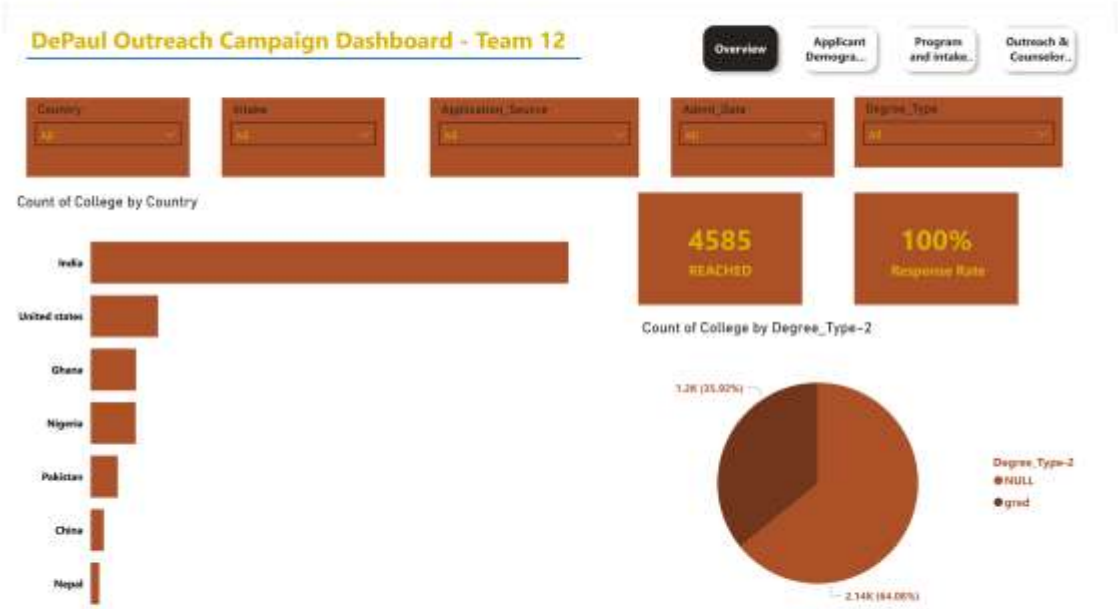
The dashboard was designed to be intuitive for both technical and non-technical users, requiring clear labeling and logical organization.

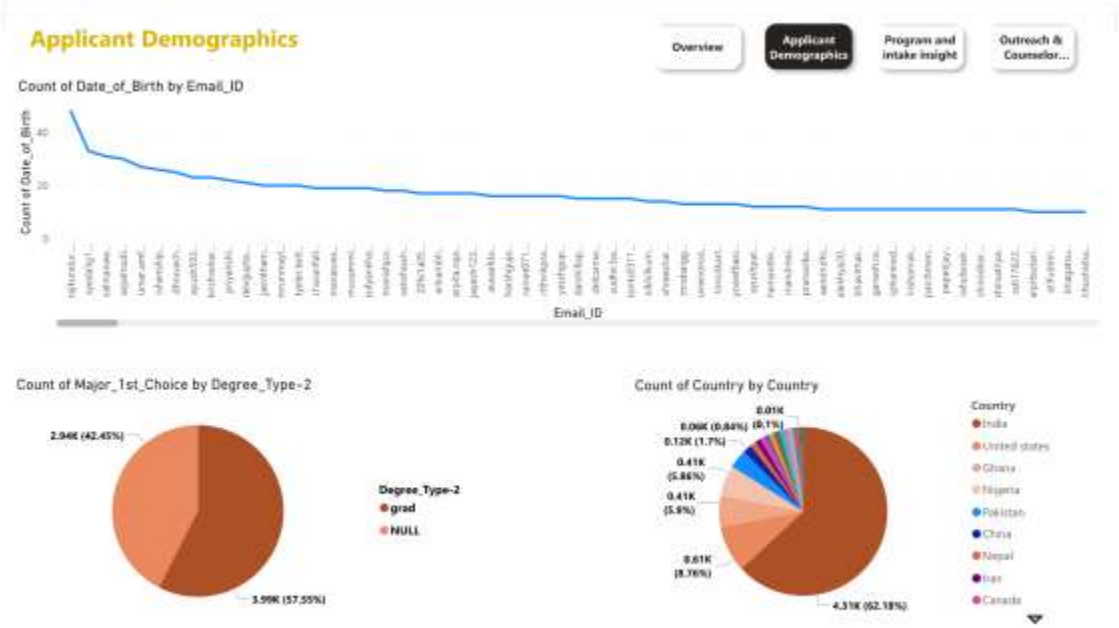
Conclusion

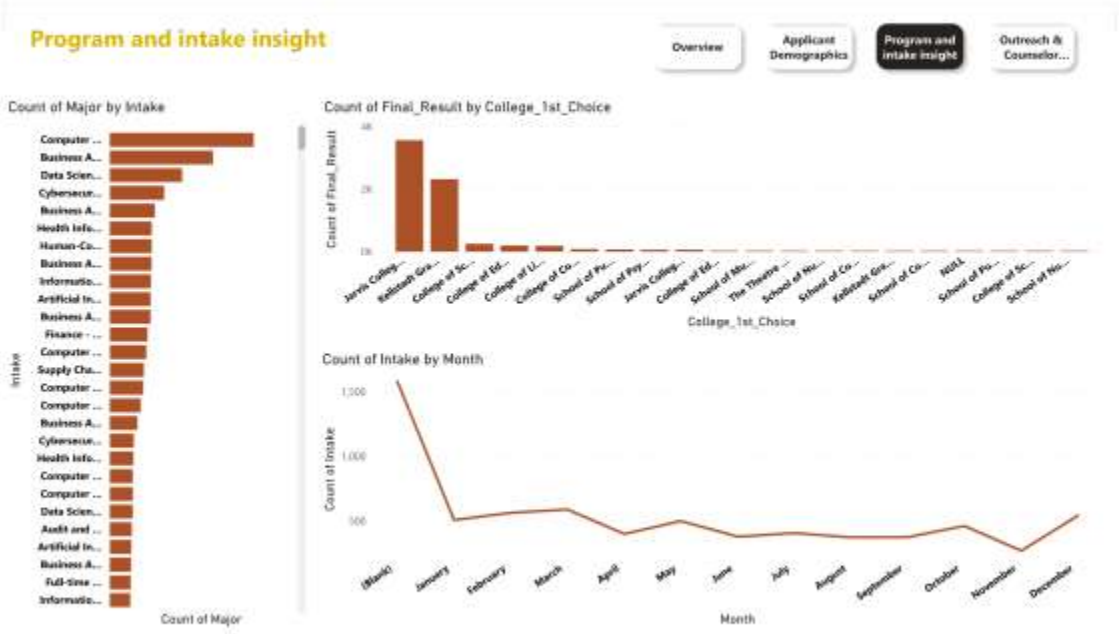
The DePaul Outreach Dashboard successfully transforms raw outreach data into a structured, interactive, and insightful reporting tool. It enables stakeholders to monitor performance, identify trends, and make informed strategic decisions.

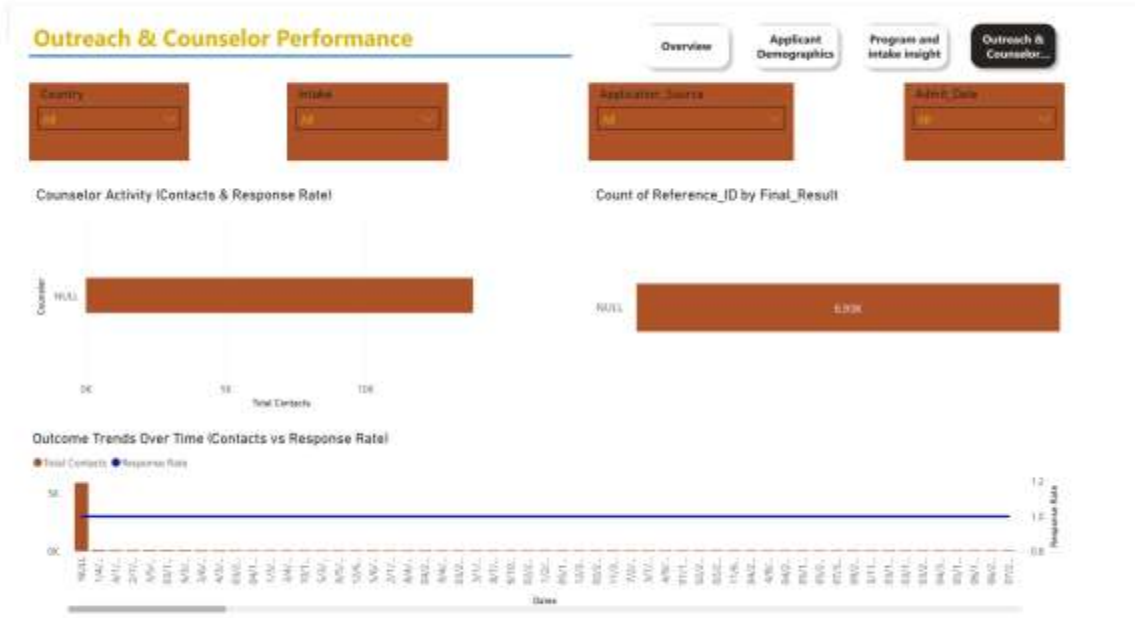
Through careful data preparation, thoughtful design, and effective visualization techniques, the dashboard provides both high-level summaries and detailed insights, supporting continuous improvement in outreach efforts.

4. Dashboard Screenshots




Applicant Demographics
Overview
Applicant Demographics
Program and intake insight
Outreach & Counselor...
Count of Date_of_Birth by Email_ID
Count of Date_of_Birth
Email_ID
Count of Major_1st_Choice by Degree_Type-2
2.94K (42.45%)
3.99K (57.55%)
Degree_Type-2
grad
NULL
Count of Country by Country
0.01K (0.1%)
0.06K (0.84%)
0.12K (1.7%)
0.41K (5.86%)
0.41K (5.9%)
0.61K (8.76%)
4.31K (62.18%)
Country
India
United states
Ghana
Nigeria
Pakistan
China
Nepal
Iran
Canada


Program and intake insight
Overview
Applicant Demographics
Program and intake insight
Outreach & Counselor...
Count of Major by Intake
Intake
Count of Major
Count of Final_Result by College_1st_Choice
Count of Final_Result
College_1st_Choice
Count of Intake by Month
Count of Intake
Month
(Blank)
January
February
March
April
May
June
July
August
September
October
November
December


Outreach & Counselor Performance
Overview
Applicant Demographics
Program and intake insight
Outreach & Counselor...
Country
Intake
Applicant_Source
Admit_Date
Counselor Activity (Contacts & Response Rate)
Counselor
NULL
Total Contacts
Count of Reference_ID by Final_Result
NULL
Outcome Trends Over Time (Contacts vs Response Rate)
Response Rate
Dates